Jaideep Manupati

+1-703-727-5315 | jaideepm21@vt.edu | jaideepmanupati21@gmail.com
linkedin.com/in/jaideep-manupati | github.com/jaideep1021

Education

Virginia Tech — Alexandria, VA
Master of Engineering in Computer Science and Applications — *Expected May 2027*

National Institute of Technology (NIT) — Srinagar, India
Bachelor of Technology in Information Technology — *July 2025*

Technical Skills

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, C, Bash

Backend & Full-Stack: FastAPI, Flask, Node.js, React, Next.js, REST APIs, SQLAlchemy, JWT, Tailwind CSS

Databases & Data Systems: PostgreSQL, TimescaleDB, pgvector, MySQL, Alembic, Vector Search, Time-Series Data

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, Sentence-Transformers, DeBERTa-v3, LangChain, RAG, BM25, NLP, LLM Evaluation

Cloud & Infrastructure: AWS (EC2, S3), Docker, Kubernetes, CI/CD, Linux, Git, Pytest

Projects

FalsePay: HCP Payments Risk & Audit Platform | Winner – Impiricus Track, VTHacks 14 — Sep 2026

- Architected the **FastAPI** backend and asynchronous **SQLAlchemy/PostgreSQL** data layer for an HCP payments risk platform analyzing $14B in annual CMS Open Payments transactions; modeled time-series payment data using **TimescaleDB hypertables** for efficient time-bucketed querying and anomaly analysis.
- Engineered an explainable risk-detection pipeline combining **Isolation Forest** anomaly detection with **pgvector** semantic-similarity search to identify specialty-to-product payment mismatches, exposed through a versioned **REST API** for downstream workflows.
- Built a secure physician audit workflow using **hashed, expiring, revocable tokens**, live CMS NPPES Registry enrichment, and AI-generated explanations; containerized application services with **Docker** for reproducible deployment.

Confid.AI: Hybrid Retrieval & LLM Evaluation Platform | Mentored by Boeing — Aug 2025 – May 2026

- Architected a modular **AI evaluation platform** using **FastAPI, LangChain, and LLaMA 3.3 70B**, with interchangeable retrieval, scoring, and inference components deployed using **Docker and Kubernetes** on Virginia Tech CS Launch infrastructure.
- Engineered a three-stage hybrid retrieval pipeline combining **Sentence-Transformer dense retrieval**, **BM25 lexical search**, and **DeBERTa-v3 cross-encoder re-ranking** over 512-token chunks, reducing hallucination rate by **40%** versus single-metric baselines.
- Developed a confidence-scoring framework measuring semantic consistency, factual completeness, and retrieval precision across grounded passages; served evaluation workflows through a **JWT-secured REST API** supporting **50+ concurrent users**.

AI-Powered Video Summarization Platform | *Flask, React, OpenAI Whisper, NLP* — Nov 2024 – Jun 2025

- Built an end-to-end **full-stack application** with **Flask, React, and OpenAI Whisper**, integrating video ingestion, transcription, NLP processing, REST APIs, and an interactive frontend for structured video summarization.
- Parallelized transcription and NLP processing to reduce end-to-end inference latency by **60%**; implemented topic segmentation and extractive-abstractive generation for timestamped summaries and chapter breakdowns.
- Integrated the **React frontend** with Flask REST endpoints to surface structured, timestamp-aligned summaries and chapter-level results, enabling users to navigate high-density academic and lecture content through a responsive end-to-end workflow.

Professional Experience

Undergraduate Research Intern — Dec 2023 – Feb 2024
National Institute of Technology (NIT) — *Warangal, India*

- Developed an end-to-end multilingual **text-to-speech system** using **Tacotron 2 and Python**, improving phoneme-to-audio accuracy by **15%** for low-resource languages; research published in *IJIRT*.
- Built phonetic alignment and spectrogram preprocessing pipelines in **PyTorch** to clean noisy speech datasets and improve mel-spectrogram reconstruction for natural speech synthesis.
- Managed reproducible research and experimentation workflows using **Git and Confluence**, documenting dataset preprocessing, experimental results, and implementation decisions to align ML development with structured software and research requirements.

Leadership & Achievements

1st Place, VTHacks 14 – Impiricus Track: Built FalsePay, an HCP payments risk and audit platform; led backend architecture and vector-search integration.

Operations Chair, Graduate Student Assembly – Virginia Tech — Jan 2026 – Present
Lead logistics and operational execution for graduate student events and technical initiatives at Virginia Tech's Innovation Campus.